

ASSIGNMENT NO.8: Views and Data Abstraction

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 4 course records and 10 student records.

Query-

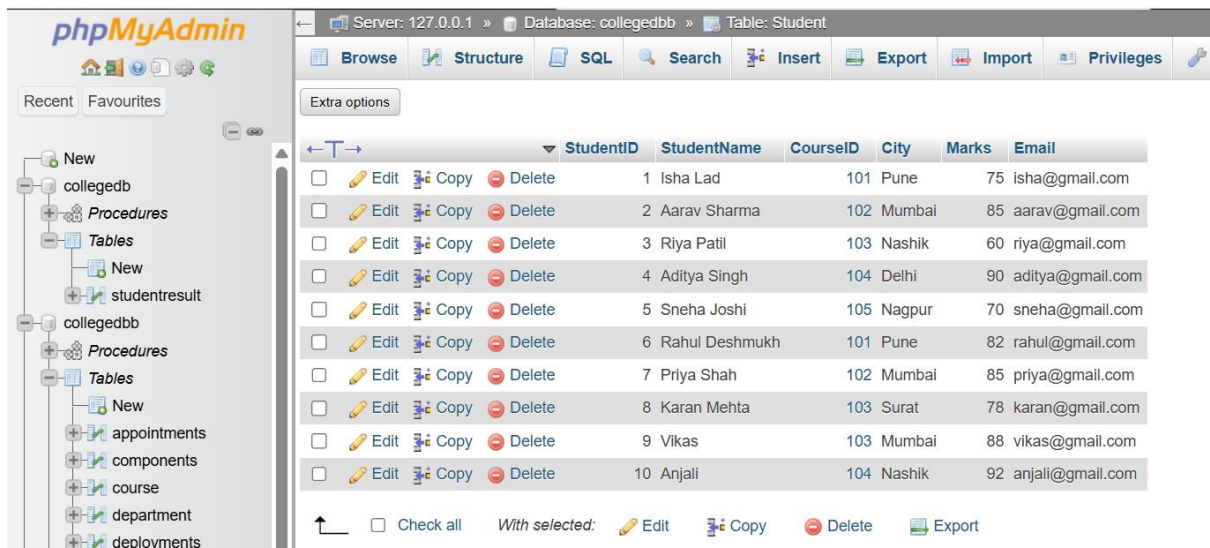
INSERT INTO Student

(StudentID, StudentName, CourseID, City, Marks, Email)

VALUES

(1, 'Rahul', 101, 'Pune', 75, 'rahul@gmail.com'),
(2, 'Priya', 102, 'Mumbai', 85, 'priya@gmail.com'),
(3, 'Amit', 103, 'Nashik', 60, 'amit@gmail.com'),
(4, 'Neha', 104, 'Pune', 90, 'neha@gmail.com'),
(5, 'Riya', 101, 'Mumbai', 70, 'riya@gmail.com'),
(6, 'Karan', 102, 'Pune', 82, 'karan@gmail.com'),
(7, 'Aarav', 101, 'Nashik', 85, 'aarav@gmail.com'),
(8, 'Sneha', 102, 'Pune', 78, 'sneha@gmail.com'),
(9, 'Vikas', 103, 'Mumbai', 88, 'vikas@gmail.com'),
(10, 'Anjali', 104, 'Nashik', 92, 'anjali@gmail.com');

Output-



	StudentID	StudentName	CourseID	City	Marks	Email
☐	1	Isha Lad	101	Pune	75	isha@gmail.com
☐	2	Aarav Sharma	102	Mumbai	85	aarav@gmail.com
☐	3	Riya Patil	103	Nashik	60	riya@gmail.com
☐	4	Aditya Singh	104	Delhi	90	aditya@gmail.com
☐	5	Sneha Joshi	105	Nagpur	70	sneha@gmail.com
☐	6	Rahul Deshmukh	101	Pune	82	rahul@gmail.com
☐	7	Priya Shah	102	Mumbai	85	priya@gmail.com
☐	8	Karan Mehta	103	Surat	78	karan@gmail.com
☐	9	Vikas	103	Mumbai	88	vikas@gmail.com
☐	10	Anjali	104	Nashik	92	anjali@gmail.com

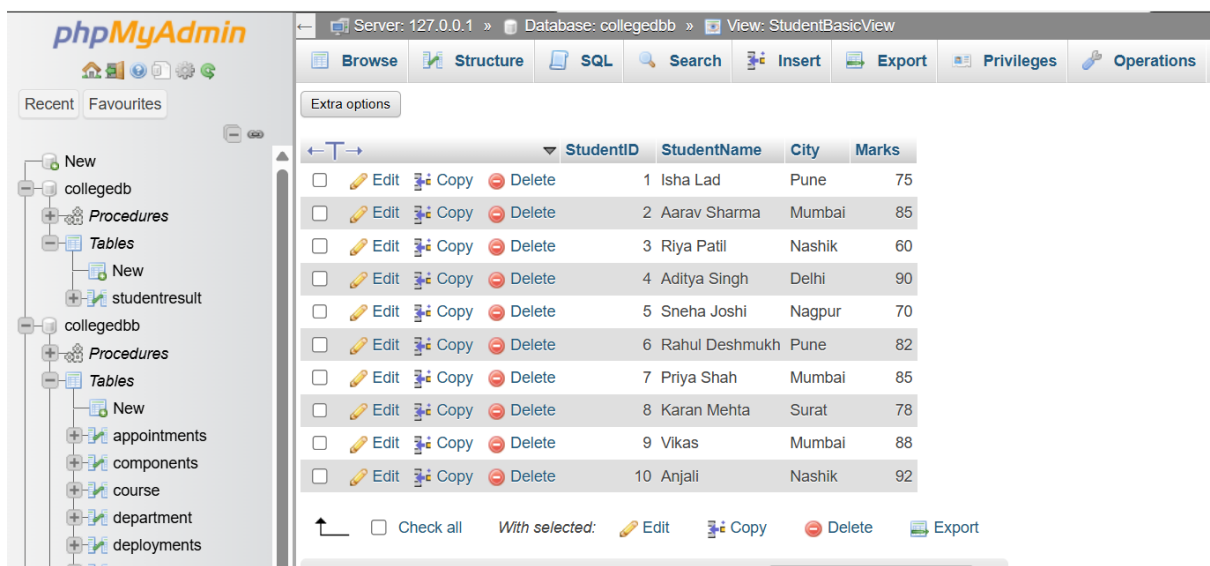
Q2) Create a view named StudentBasicView to display only selected student details.
The view should contain:

Column
StudentID
StudentName
City
Marks

Query-

```
CREATE VIEW StudentBasicView AS  
SELECT StudentID, StudentName, City, Marks  
FROM Student;  
SELECT * FROM StudentBasicView;
```

Output-



	StudentID	StudentName	City	Marks
☐	1	Isha Lad	Pune	75
☐	2	Aarav Sharma	Mumbai	85
☐	3	Riya Patil	Nashik	60
☐	4	Aditya Singh	Delhi	90
☐	5	Sneha Joshi	Nagpur	70
☐	6	Rahul Deshmukh	Pune	82
☐	7	Priya Shah	Mumbai	85
☐	8	Karan Mehta	Surat	78
☐	9	Vikas	Mumbai	88
☐	10	Anjali	Nashik	92

Q3) Create a view named StudentCourseView using Student and Course tables.

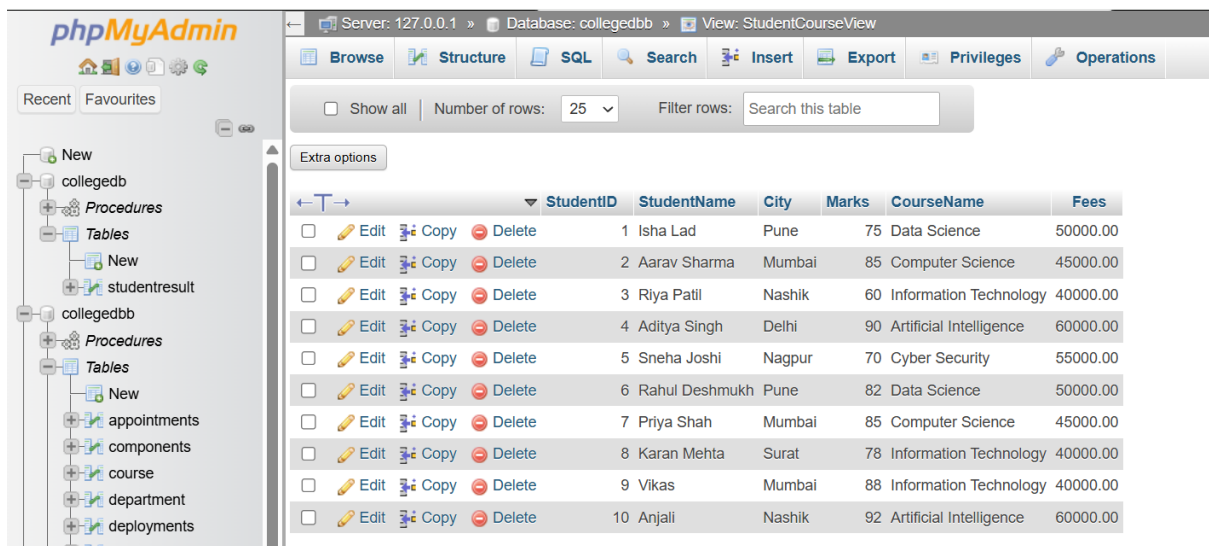
The view should display:

Column
StudentID
StudentName
City
Marks
CourseName
Fees

Query-

```
CREATE VIEW StudentCourseView AS
SELECT
Student.StudentID,
Student.StudentName,
Student.City,
Student.Marks,
Course.CourseName,
Course.Fees
FROM Student
INNER JOIN Course
ON Student.CourseID = Course.CourseID;
```

Output-



The screenshot shows the phpMyAdmin interface with the 'StudentCourseView' view selected. The view displays 10 rows of student data. The columns are StudentID, StudentName, City, Marks, CourseName, and Fees. The data is as follows:

StudentID	StudentName	City	Marks	CourseName	Fees
1	Isha Lad	Pune	75	Data Science	50000.00
2	Aarav Sharma	Mumbai	85	Computer Science	45000.00
3	Riya Patil	Nashik	60	Information Technology	40000.00
4	Aditya Singh	Delhi	90	Artificial Intelligence	60000.00
5	Sneha Joshi	Nagpur	70	Cyber Security	55000.00
6	Rahul Deshmukh	Pune	82	Data Science	50000.00
7	Priya Shah	Mumbai	85	Computer Science	45000.00
8	Karan Mehta	Surat	78	Information Technology	40000.00
9	Vikas	Mumbai	88	Information Technology	40000.00
10	Anjali	Nashik	92	Artificial Intelligence	60000.00

Q4) Write a MySQL query using StudentCourseView to display students from Mumbai.

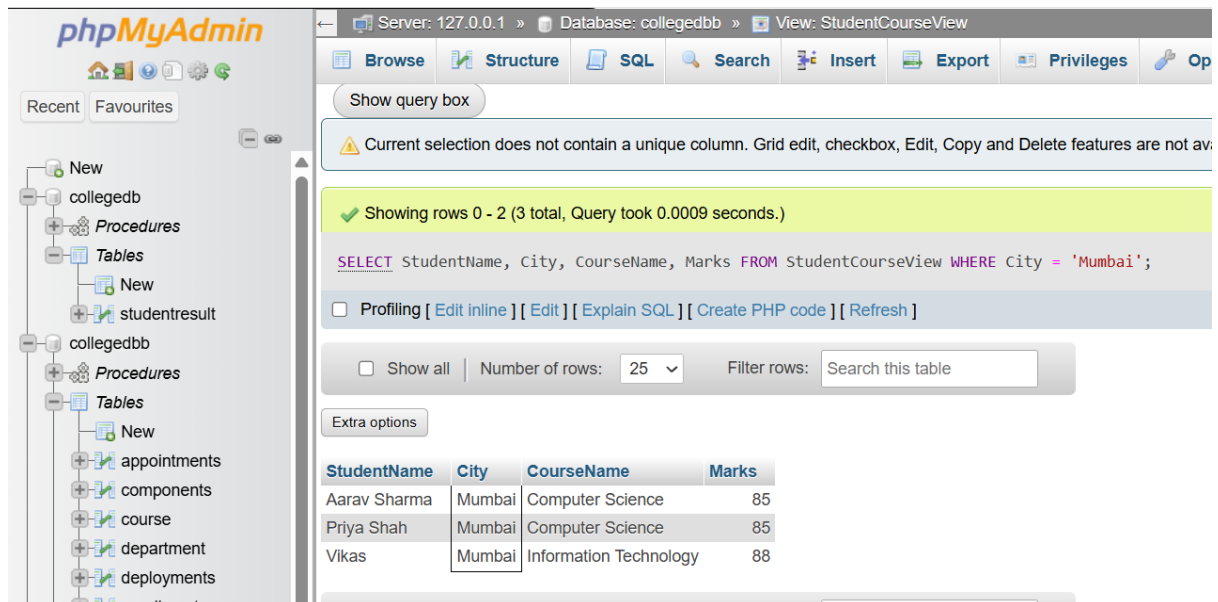
The output should contain:

Column
StudentName
City
CourseName
Marks

Query-

```
SELECT StudentName, City, CourseName, Marks  
FROM StudentCourseView  
WHERE City = 'Mumbai';
```

Output-



The screenshot shows the phpMyAdmin interface. The left sidebar displays the database structure for 'collegedbb'. The main panel shows the 'StudentCourseView' view. A warning message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green status bar indicates 'Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)'. The SQL query is displayed in the query box: `SELECT StudentName, City, CourseName, Marks FROM StudentCourseView WHERE City = 'Mumbai';`. Below the query box, there are options for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A 'Show all' checkbox is present, along with a 'Number of rows' dropdown set to 25 and a 'Filter rows' search box. An 'Extra options' button is also visible. The result table is displayed below, showing 3 rows of data.

StudentName	City	CourseName	Marks
Aarav Sharma	Mumbai	Computer Science	85
Priya Shah	Mumbai	Computer Science	85
Vikas	Mumbai	Information Technology	88

Q5) Write a MySQL query using StudentCourseView to display students sorted by marks in descending order.

The output should contain:

Column
StudentName
CourseName
Marks

Query-

```
SELECT StudentName, CourseName, Marks  
FROM StudentCourseView  
ORDER BY Marks DESC
```

Output-

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'collegeddb' selected. The main panel displays the 'StudentCourseView' table. A status bar at the top indicates 'Showing rows 0 - 9 (10 total, Query took 0.0023 seconds.) [Marks: 92... - 60...]'. Below this, the SQL query is shown: `SELECT StudentName, CourseName, Marks FROM StudentCourseView ORDER BY Marks DESC;`. There are options for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A 'Show all' checkbox is present, along with a 'Number of rows' dropdown set to 25 and a 'Filter rows' search box. An 'Extra options' button is also visible. The table data is displayed with columns 'StudentName', 'CourseName', and 'Marks'. The rows are sorted by marks in descending order.

StudentName	CourseName	Marks
Anjali	Artificial Intelligence	92
Aditya Singh	Artificial Intelligence	90
Vikas	Information Technology	88
Aarav Sharma	Computer Science	85
Priya Shah	Computer Science	85
Rahul Deshmukh	Data Science	82
Karan Mehta	Information Technology	78
Isha Lad	Data Science	75
Sneha Joshi	Cyber Security	70
Riya Patil	Information Technology	60

Q6) Create a report view named ManagementReportView.

The view should display course-wise summary:

Column
CourseName
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Also write one line explaining why such a view is useful in reporting.

Query-

```
CREATE VIEW ManagementReportView AS
SELECT
Course.CourseName,
COUNT(Student.StudentID) AS TotalStudents,
AVG(Student.Marks) AS AverageMarks,
MIN(Student.Marks) AS LowestMarks,
MAX(Student.Marks) AS HighestMarks
FROM Course
LEFT JOIN Student
ON Course.CourseID = Student.CourseID
GROUP BY Course.CourseName;
SELECT * FROM ManagementReportView;
```

Output-

The screenshot shows the phpMyAdmin interface with the 'collegedbb' database selected. The 'Tables' tab is active, displaying a list of tables including 'appointments', 'components', 'course', 'department', 'deployments', and 'enrollment'. The 'ManagementReportView' table is selected, and its structure is shown. The table has five columns: 'CourseName', 'TotalStudents', 'AverageMarks', 'LowestMarks', and 'HighestMarks'. The data is displayed in a table format with 6 rows. The first row is 'Artificial Intelligence' with 2 students, an average mark of 91.0000, and marks ranging from 90 to 92. The second row is 'BCA' with 0 students and NULL values for marks. The third row is 'Computer Science' with 2 students, an average mark of 85.0000, and marks ranging from 85 to 85. The fourth row is 'Cyber Security' with 1 student, an average mark of 70.0000, and marks ranging from 70 to 70. The fifth row is 'Data Science' with 2 students, an average mark of 78.5000, and marks ranging from 75 to 82. The sixth row is 'Information Technology' with 3 students, an average mark of 75.3333, and marks ranging from 60 to 88.

Server: 127.0.0.1 » Database: collegedbb » View: ManagementReportView

Showing rows 0 - 5 (6 total, Query took 0.0011 seconds.)

`SELECT * FROM ManagementReportView;`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

CourseName	TotalStudents	AverageMarks	LowestMarks	HighestMarks
Artificial Intelligence	2	91.0000	90	92
BCA	0	NULL	NULL	NULL
Computer Science	2	85.0000	85	85
Cyber Security	1	70.0000	70	70
Data Science	2	78.5000	75	82
Information Technology	3	75.3333	60	88

☐ Show all | Number of rows: 25 | Filter rows:

ManagementReportView is useful because it provides a summarized course-wise report of student count and marks, making data analysis and reporting easier.